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A [1]H NMR-based approach to investigate metabolomic differences in the plasma and urine of young women after cranberry juice or apple juice consumption

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The overall metabolic changes caused by cranberry juice or apple juice consumption using a global [1]H NMR-based metabolomics approach were investigated. Eighteen female college students were given either cranberry or apple juice for three days using a cross-over design. Plasma and urine samples were collected and analyzed using [1]H NMR-based metabolomics followed by multivariate analyses. No metabolic difference was observed in plasma before and after juice consumption. However, metabolome in plasma and urine after cranberry juice consumption were different from those after apple juice consumption. Cranberry juice consumption caused a greater increase in urinary excretion of hippuric acid and a higher level of citric acid in the plasma. Furthermore, cranberry juice decreased the plasma level of lactate, D-glucose, and two unidentified metabolites compared to apple juice consumption. The metabolomic changes caused by cranberry juice consumption may help to explain its reported health benefits.

1. Introduction

Cranberries (*Vaccinium macrocarpon*) are a native crop in North America grown commercially in Wisconsin, Massachusetts, New Jersey, Washington, and part of Canada. Fresh cranberries have a tart taste, therefore majority of them are processed into juice for consumption. Cranberries are a rich source of phytochemicals including anthocyanins, flavan-3-ols, and proanthocyanidins (Prior, Lazarus, Cao, Muccitelli, & Hammerstone, 2001; Yan, Murphy, Hammond, Vinson, & Neto, 2002). Proanthocyanidins are oligomeric or polymeric of

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Abbreviations: CPMG, Carr–Purcell–Meiboom–Gill; DP, degree of polymerization; DSS, sodium 2,2-dimethyl-2-silapentane-5-sulfonate; NMR, nuclear magnetic resonance; NOESY, Nuclear Overhauser effect spectroscopy; OPLS-DA, orthogonal projection on latent structure-discriminant analysis; PCA, principal component analysis; PLS-DA, projection on latent structure-discriminant analysis

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flavan-3-ols linked through interflavan bonds. B-type interflavan linkage is C4 → C8 and/or C4 → C86. A-type proanthocyanidins contain an additional ether bond between C2 and C7 (Ou & Gu, 2014). Most foods including apples or apple juice contain exclusively B-type proanthocyanidins, while cranberries or cranberry juice contain both A- and B-type proanthocyanidins (Gu et al., 2003). Ingestion of cranberry juice has long been associated with prevention of urinary tract infection (UTI) (Blatherwick, 1914). Studies showed that A-type proanthocyanidins from cranberry juice inhibited the adhesion of uropathogenic *Escherichia coli*, whereas B-type proanthocyanidins from apple juice showed no activity (Howell et al., 2005). Anti-adhesion activity in human urine was detected following cranberry juice cocktail consumption, but not after consumption of the apple juice (Howell et al., 2005). A-type dimers and trimers demonstrated anti-adhesion activity, whereas epicatechin and a B-type dimer showed no such effect (Foo, Lu, Howell, & Vorsa, 2000).

Metabolomics refers to the comprehensive analysis of small molecular-weight metabolites present in biological samples (Nicholson, Lindon, & Holmes, 1999). It is a powerful platform to monitor the metabolic flux following genetic modification, pathophysiological changes, or exogenous challenges (Griffin, 2006; Nicholson et al., 1999). Diet is considered an important factor to impact human metabolome. Part of the ingested phytochemicals from foods are absorbed through gut barrier and metabolized. The resultant exogenous phytochemical metabolites are part of food metabolome (Manach, Hubert, Llorach, & Scalbert, 2009) and may alter the endogenous metabolites. For instance, cranberries are known to impact gene expression, protein activity, and signaling transduction (Deziel, Patel, Neto, Gottschall-Pass, & Hurta, 2010; Kresty, Howell, & Baird, 2011). Cranberry juice consumption may alter the profile of endogenous metabolites. Alteration of endogenous metabolites is considered as the amplified ‘end-point’ output of changes down the biochemical pathways (Lin, Chan, Li, & Cai, 2010). Metabolomics in humans is expected to shed a light on the physiological effects of phytochemical intake and identify new candidate biomarkers. Multiple analytical platforms are employed to efficiently generate the metabolic profiles of biological samples. Nuclear magnetic resonance (NMR) spectroscopy is one of the most commonly used analytic techniques. It is able to detect a wide range of metabolites in biological fluids. NMR spectroscopy has the advantages of being quantitative, highly reproducible, non-selective and minimal sample preparation (Dunn, Broadhurst, Atherton, Goodacre, & Griffin, 2011). Metabolomics strategy produces high-dimensional and complex data set. Multivariate statistic techniques including PCA, PLS-DA, or OPLS-DA are often used to reduce the dimensionality of the data (Bylesjö et al., 2006). They are useful to reveal patterns related to the physiological or pathological perturbation and to aid biological interpretation (Marchesi et al., 2007).

We hypothesized that cranberry juice consumption may have a different impact on human metabolome compared to apple juice. A ^{1}H NMR-based metabolomics approach with multivariate statistic techniques was applied to analyze the overall metabolic impact of cranberry juice and differentiate that from apple juice consumption.

2. Materials and methods

2.1. Chemicals and materials

Cranberry juice cocktail (double strength, 54% juice) and 100% apple juice were provided by Ocean Spray Cranberries, Inc. (Lakeville-Middleboro, MA, USA). Gallic acid, HPLC-grade acetonitrile, methylene chloride, methanol, acetic acid, Folin–Ciocalteu reagent, sodium carbonate, sodium phosphate dibasic anhydrous, sodium phosphate monobasic anhydrous, sodium hydroxide, sodium chloride, sodium azide, sucrose, glucose, and fructose were purchased from Fischer Scientific Co. (Pittsburgh, PA, USA). D_2O (99.9% D), 2, 2-dimethyl-2-silapentane-5-sulfonate (DSS, 98%) were from Cambridge Isotope Laboratories, Inc. Sephadex LH-20 resin was purchased from Sigma-Aldrich (St. Louis, MO, USA). Amberlite FPX 66 resin was a product of Rohm and Haas Co. (Philadelphia, PA, USA). Pooled plasma used as quality control samples were purchased from American Red Cross and they were collected over a period of about 2 weeks.

2.2. Total phenolics, total anthocyanins, proanthocyanidin composition and content

Two hundred and fifty milliliters of cranberry juice or apple juice were loaded onto a column packed with Amberlite FPX 66 resins. Column was eluted with 3 l of water to remove sugars and ascorbic acids. Column was eluted with 300–400 ml of methanol to yield cranberry juice sugar-free extract (890 mg) or apple juice sugar-free extract (393 mg). The total phenolic content of juice sugar-free extracts were determined by Folin–Ciocalteu assay (Singleton & Rossi, 1965). A pH differential assay was used to determine the total anthocyanin content (Giusti & Wrolstad, 2001). Another 250 ml of cranberry juice or apple juice was loaded onto a column (5.8 × 28 cm) packed with Sephadex LH-20. The column was eluted with 30% methanol to remove anthocyanins and phenolic acids, and then eluted with 70% acetone to yield cranberry juice proanthocyanidins extract (273 mg) or apple juice proanthocyanidins extract (18.5 mg). Then proanthocyanidin composition and content were analyzed on an Agilent 1200 HPLC system (Palo Alto, CA, USA) equipped with a binary pump, an autosampler, a fluorescence detector, and a HCT ion trap mass spectrometer (Bruker Daltonics, Billerica, MA, USA). Separation of proanthocyanidins was carried out on a 250 mm × 4.6 mm i.d., 5 μm, Develosil Diol 100 Å column, with a 4 mm × 3 mm i.d. guard column (Phenomenex, Torrance, CA, USA) at a column temperature of 35 °C. The binary mobile phase consisted of (A) acetonitrile/acetic acid (98:2, v/v) and (B) methanol/water/acetic acid (95:3:2, v/v/v). The 76 min gradient was as follows: 0–12 min, 7% B isocratic; 12–60 min, 7–37.6% B linear; 60–63 min, 37.6–100% B linear, 63–70 min 100% B isocratic; 70–76 min 100–7% B linear; followed by 5 min of re-equilibration of the column before the next run. Excitation and emission of the fluorescent detector were set at 231 and 320 nm, respectively. Electrospray ionization at negative mode was performed using nebulizer 50 psi, drying gas 10 l/min, drying temperature 350 °C, and capillary 4000 V. Mass spectra were recorded from *m/z* 150 to 2000. The most abundant ion in full scan was isolated, and its product ion spectra were recorded.

2.3. Sugar analysis in cranberry juice and apple juice

Sugar analysis was conducted on an Agilent 1200 HPLC system consisting of an autosampler, a binary pump, and a refractive index detector (Agilent Technologies, Palo Alto, CA, USA). Separation was carried out on a Restek ultra amino column (5 µm, 250 × 4.6 mm). The column temperature was maintained at 30 °C and a 5 µl of sample were injected. Acetonitrile/water (80:20, v/v) was used as the mobile phase at a constant flow rate of 1.0 ml/min. The optical unit temperature was set at 35 °C and the refractive index detector signal was monitored in positive polarity. The run time for each sample was 15 min followed by 5 min post time before the next run. Calibration curves were constructed by use of pure standards of glucose, fructose, and sucrose.

2.4. Subjects and study design

Human study was approved by Institutional Review Boards at University of Florida. Eighteen healthy female college students between 21 and 29 years old with a normal BMI of 18.5–25 were recruited. Each subject was provided with a list of foods that contained significant amount of proanthocyanidins, such as cranberries, apples, grapes, blueberries, chocolate and plums. They were advised to avoid these foods during the 1–6th day and the rest of the study. On the morning of the 7th day, a first-morning baseline urine sample and blood sample were collected from all human subjects after overnight fasting. Participants were then randomly allocated into two groups ($n = 9$) to consume cranberry juice or apple juice. Six bottles (250 ml/bottle) of juice were given to participants to drink in the morning and evening of the 7th, 8th, and 9th day. On the morning of 10th day, all subjects returned to the clinical unit to provide a first-morning urine sample after overnight fasting. The blood sample was also collected from participants 30 min later after they drank another bottle of juice in the morning. After two-weeks of wash out period, participants switched to the alternative regimen and repeated the protocol. Blood samples were centrifuged at 2000 *g* for 10 min at 4 °C to obtain plasma. All urine and plasma samples were aliquoted and kept in a −80 °C freezer until analysis. One human subject was dropped off this study because she missed part of her appointments. Another two human subjects were removed from urine metabolomics analyses because they failed to provide required urine samples after juice drinking.

2.5. ^{1}H NMR metabolomics analyses

Plasma or urine samples were taken out of −80 °C freezer to thaw at room temperature, and then centrifuged at 5220 RCF (relative centrifugal force) for 5 min. Plasma (400 µl) was mixed with 200 µl of saline solution (0.9% NaCl in D_2O). Urine (400 µl) was mixed with 200 µl of phosphate buffer (pH 7.4, DSS added). Both plasma and urine samples were transferred into 5 mm Bruker NMR tubes (Z105684 Bruker 96 well rack) using Gilson 215 Liquid Handler (Trilution software version 2.0). All ^{1}H NMR spectra were collected on a 600 MHz Avance II NMR spectrometer (Bruker Biospin, Rheinstetten, Germany) equipped with a 5 mm cryo probe. Instrument had a sample changer (Sample Xpress Lite Autosampler) under Icon-NMR. 1D NOESY-presaturated spectra for urine and 1D CPMG-presaturated spectra for plasma were recorded. All NMR data were acquired at 25 °C. Probe tuning and matching were optimized for the representative sample in each run. A 90° pulse length, the offset of the water signal, water suppression and receiver gain for a data set were also determined on the representative sample in each run. The probe was automatically locked to H_2O+D_2O (90%+10%) and shimmed for each sample.

2.6. Multivariate data processing and statistical analyses

All NMR spectra were phased and baseline corrected using NMRPipe and then converted to FT (Fourier transformed) files. The FT files were imported into MATLAB (R2013B, the Mathworks, Inc., Natick, MA, USA). Water regions (4.6–5.1 ppm) and DSS region (−0.2–0.5 ppm) were removed. Then the spectra were aligned and normalized in MATLAB. The resultant data set was imported into SIMCA (Version 13.0.3, Umetrics, Umea, Sweden) for multivariate statistical analysis. Data were Pareto scaled before PCA, PLS-DA and OPLS-DA analyses in SIMCA. Unsupervised PCA model was performed to initially examine intrinsic variation in the data set. Then supervised pattern recognition methods, projection on latent structure-discriminant analysis (PLS-DA) and orthogonal projection on latent structure-discriminant analysis (OPLS-DA) (Bylesjö et al., 2006) were used to extract maximum information on discriminant compounds from the data. Validation of the model was tested using 7-fold internal cross-validation and permutation tests for 100 times. To further evaluate the predictive ability of the PLS-DA and OPLS-DA models, an external validation procedure was performed (Brindle et al., 2002; Llorach et al., 2010). The whole data set was split into a training set and a test set. Approximately 70% of the samples were randomly selected as training set and the remaining 30% were treated as test set. PLS-DA and OPLS-DA models were built based on the training set and obtained models were used to blindly predict the classification of the samples in the test set. This procedure was repeated 30 times and correct classification rate was calculated. For univariate analyses, Welch's *t* test was carried out on the NMR signal intensity of selected metabolites which were considered to be responsible for the separation between treatments from multivariate analyses. Benjamini and Hochberg (1995) procedure ($\alpha = 0.01$) was conducted to control false discoveries. Box-and-whisker plot was drawn to display variations in samples within each treatment. The univariate analyses were done using Excel Microsoft (Version 2007, Microsoft Corporation., Seattle, WA, USA).

3. Results and discussion

3.1. Juice analyses

Fig. S1 in the supplementary data shows the HPLC chromatograms of proanthocyanidins extracted from cranberry juice and apple juice using fluorescence detection. Both A-type and B-type

Table 1 – Proanthocyanidins, total phenolics, total anthocyanins and sugar content of cranberry juice and apple juice.

	Cranberry juice	Apple juice
Proanthocyanidin content (μg/ml juice)		
Monomer	6.39 ± 0.19	Not detected
Dimers	53.8 ± 0.1	0.225 ± 0.170
Trimers	49.2 ± 0.7	0.445 ± 0.012
Tetramers	58.5 ± 0.5	1.26 ± 0.07
Pentamers	34.4 ± 1.8	1.28 ± 0.01
High polymers	364 ± 14	6.46 ± 0.41
Total	566 ± 17	9.68 ± 0.52
Total phenolics (μg gallic acid equivalents/ml juice)*		
Total phenolics	913 ± 7	124 ± 1
Total anthocyanins (μg cyanidin 3,5-diglucoside equivalents/ml juice)		
Total anthocyanins	59.2 ± 2.4	0.12 ± 0.00
Sugar composition and content (mg/ml juice)		
Fructose	3.46 ± 0.12	157 ± 8
Glucose	22.3 ± 0.3	76.1 ± 2.3
Sucrose	Not detected	41.6 ± 1.5
Total	25.8 ± 0.4	275 ± 12

Data are expressed as mean ± standard deviation of duplicate tests.
* Ascorbic acid was not counted as total phenolics.

proanthocyanidins were detected in cranberry juice, while only B-type proanthocyanidins and their oxidized forms were found in apple juice. Oligomeric proanthocyanidins with degree of polymerization (DP) 1–5 and high polymeric proanthocyanidins in juice were identified and quantified by HPLC-MS^n. The total quantifiable proanthocyanidins content was 566 μg/ml in cranberry juice and 9.68 μg/ml in apple juice. The total phenolics and total anthocyanins of apple juice (124 μg gallic acid/ml, 0.12 μg cyanidin 3,5-diglucoside/ml) were lower than those in cranberry juice (913 μg gallic acid/ml, 59.2 μg cyanidin 3,5-diglucoside/ml) (Table 1). It should be noted that ascorbic acid was removed using a chromatographic method so it was not counted as part of total phenolics. Individual phenolics and anthocyanins were not quantified since they were not the targeted phytochemicals in this study. Previous studies have found phenolic acids including *p*-hydroxybenzoic acid, *p*-coumaric acid, ferulic acid, and anthocyanins such as cyanidin-3-O-glucoside, pelargonidin-3-O-glucoside and cyanidin were in cranberry seeds (Chen et al., 2014). Phloretin, phenolic acids, kaempferol, myricetin, quercetin, cyanidin, delphinidin and peonidin were identified in apples (Zhao, Bomser, Joseph, & DiSilvestro, 2013).

Sugar composition and content were analyzed on HPLC using a refractive index detector. Glucose, fructose and sucrose were found in apple juice, while only glucose and fructose appeared in cranberry juice (Fig. S2). The results were consistent with previously reported findings (Fuleki, Pelayo, & Palabay, 1994). Table 1 shows that total sugar content in apple juice was 10 times higher than that in cranberry juice, with fructose being the predominant one.

3.2. *Quality control data*

To validate NMR acquisition method, quality control samples consisting of 17 replicates of pooled plasma were analyzed along with experimental samples. PCA model was built to analyze the metabolic differences between quality control and experimental samples. The mechanism was based on the ability of PCA model to cluster samples in an unsupervised approach. PCA score plot (Fig. S3) shows that the 17 replicates were tightly clustered and segregated from plasma samples from this study, suggesting that our data acquisition method was valid.

3.3. *Multivariate analyses of plasma after drinking cranberry juice* vs. *drinking apple juice*

PCA, PLS-DA and OPLS-DA models were built to analyze the metabolic patterns of plasma and urine. Neither PCA nor OPLS-DA models were able to detect metabolic differences in human plasma of baseline samples and samples after cranberry or apple juice consumption (data not shown). However, overall metabolic profiles of plasma and urine after cranberry juice consumption were found to be different from those after apple juice consumption. Fig. 1A and B showed that two groups of plasma samples were segregated on the score plot of OPLS-DA model but not on PCA model. Compared with PLS-DA model, OPLS-DA performed the analysis with orthogonal filtration of matrix X on a vector Y. The variance in the X matrix was split by OPLS model into predictive and orthogonal variances. “Structure noise” of data matrix which was unrelated to the variation of interest such as genetic background, age, physical activity, stress, etc., was filtered and described only by the orthogonal components. The variation of scientific interest was only observed in the predictive component, which is the first component. Therefore the interpretability of the resulting model was increased (Fonville et al., 2010). Plasma after drinking cranberry juice and apple juice were clearly segregated on the score plot of OPLS-DA. One predictive component and seven orthogonal components were generated in the OPLS-DA. This model obtained high-quality parameters with an overall value of R^2X, R^2Y, and Q^2 of 0.856, 0.979, and 0.652, respectively (Table 2). It indicated that about 85% of variance in X data matrix and 98% of variance in Y data set was explained by this model. Model validation was performed by permutation test ($n = 100$) and 7-fold internal validation. Calculated Q^2 of 0.652 from cross-validation was higher than 0.5, which was considered as good for metabolomics data. High Q^2 indicated that the OPLS-DA model had a good predictability. The Q^2-intercept values from permutation test (Fig. 2A) were lower than 0.05, indicating that the achieved segregation was not due to overfitting. OPLS-DA score plot and cross-validated score plot were visualized in Fig. 3. Although one plasma sample after cranberry juice consumption and two plasma samples after apple juice consumption were misclassified during internal validation, the rest of samples were correctly classified into two groups.

3.4. *Multivariate analyses of urine after drinking cranberry juice* vs. *drinking apple juice*

Similar results were obtained from urine metabolomics data. As an unsupervised technique, PCA reveals the main structure in the data without considering a special direction or type of information. The score plot of PCA in Fig. 1C shows some segregation between two groups of urine samples. By using supervised pattern recognition techniques, a much clearer

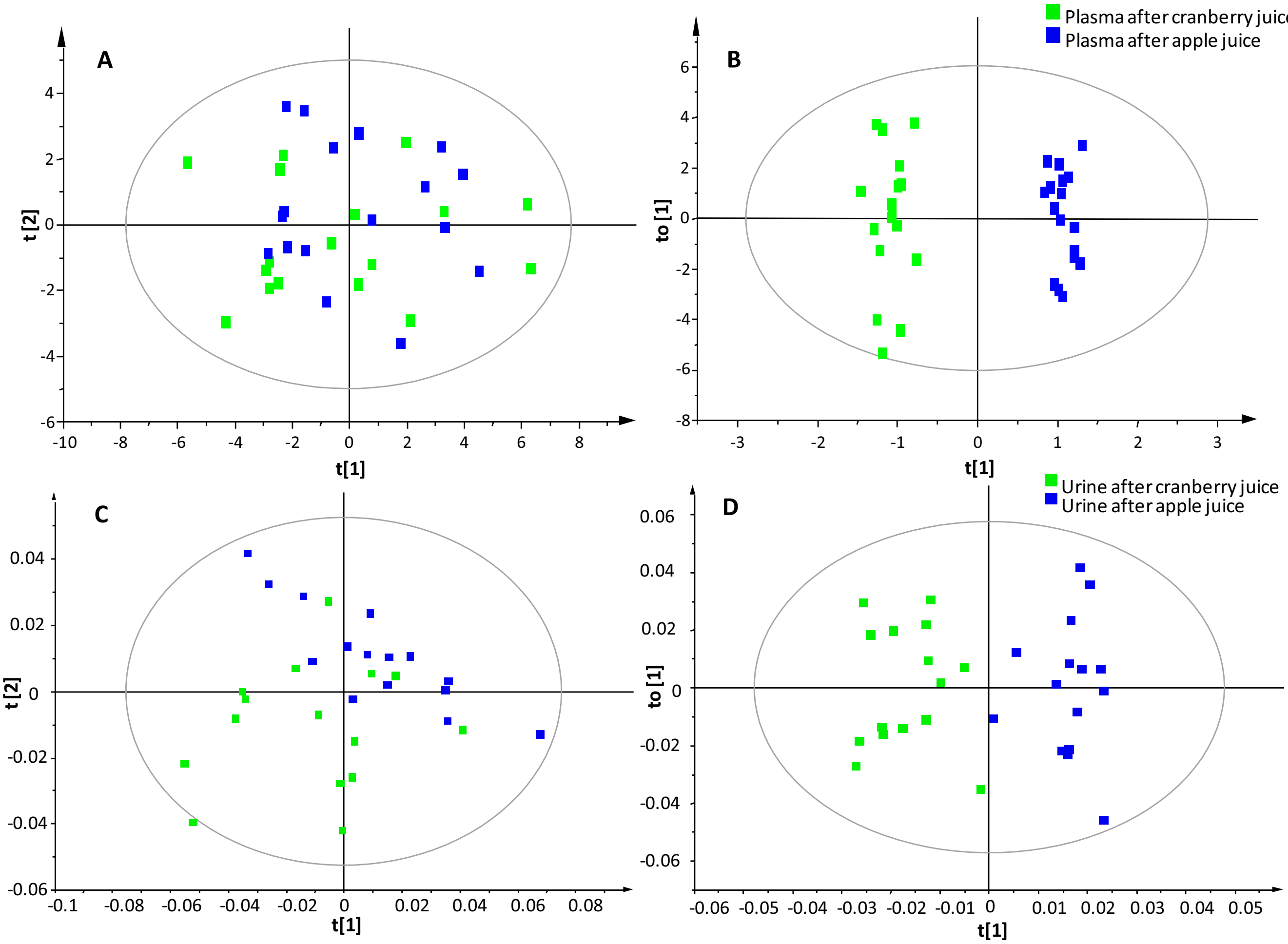

Fig. 1 – The PCA (A), OPLS-DA (B) score plot of plasma, and PCA (C), OPLS-DA (D) score plot of urine following juice drinking. Green squares: plasma or urine after cranberry juice. Blue squares: plasma or urine after apple juice. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

segregation was observed on the score plot of PLS-DA (data not shown) and OPLS-DA (Fig. 1D). One predictive component and two orthogonal components were generated from the OPLS-DA model. An overall value of R^2X, R^2Y, and Q^2 of 0.548, 0.853, and 0.503 indicated high quality of the OPLS-DA model (Table 2). Model validation was performed by permutation test (n = 100) and 7-fold internal validation. Q^2 of 0.503 suggested that the model had a good predictive ability. The regression line generated from permutation test (Fig. 2B) suggested that the OPLS-DA model was valid. The cross-validated score plot (Fig. 4) was

Table 2 – Summary of parameters for PCA, PLS-DA, and OPLS-DA models for plasma and urine after drinking cranberry juice and after drinking apple juice.

	Model for plasma			Model for urine		
	PCA	PLS-DA	OPLS-DA	PCA	PLS-DA	OPLS-DA
N[a]	3	3	1P[c]+7O[d]	2	3	1P[c]+2O[d]
R^2X(cum)[b]	0.683	0.571	0.856	0.505	0.548	0.548
R^2Y(cum)[b]	–	0.716	0.979	–	0.853	0.853
Q^2(cum)[b]	0.521	0.414	0.652	0.414	0.547	0.503
Correct classification rate*	–	0.803 ± 0.098	0.803 ± 0.091	–	0.802 ± 0.108	0.802 ± 0.101

[a] N: number of components.
[b] R^2X (cum) and R^2Y (cum) are the cumulative modeled variations in the X and Y matrix, respectively. Q^2Y (cum) is the cumulative predicted variation in the Y matrix.
[c] Predictive component.
[d] Orthogonal component.
* Correct classification rate was obtained from external validation procedure repeated for 30 times.

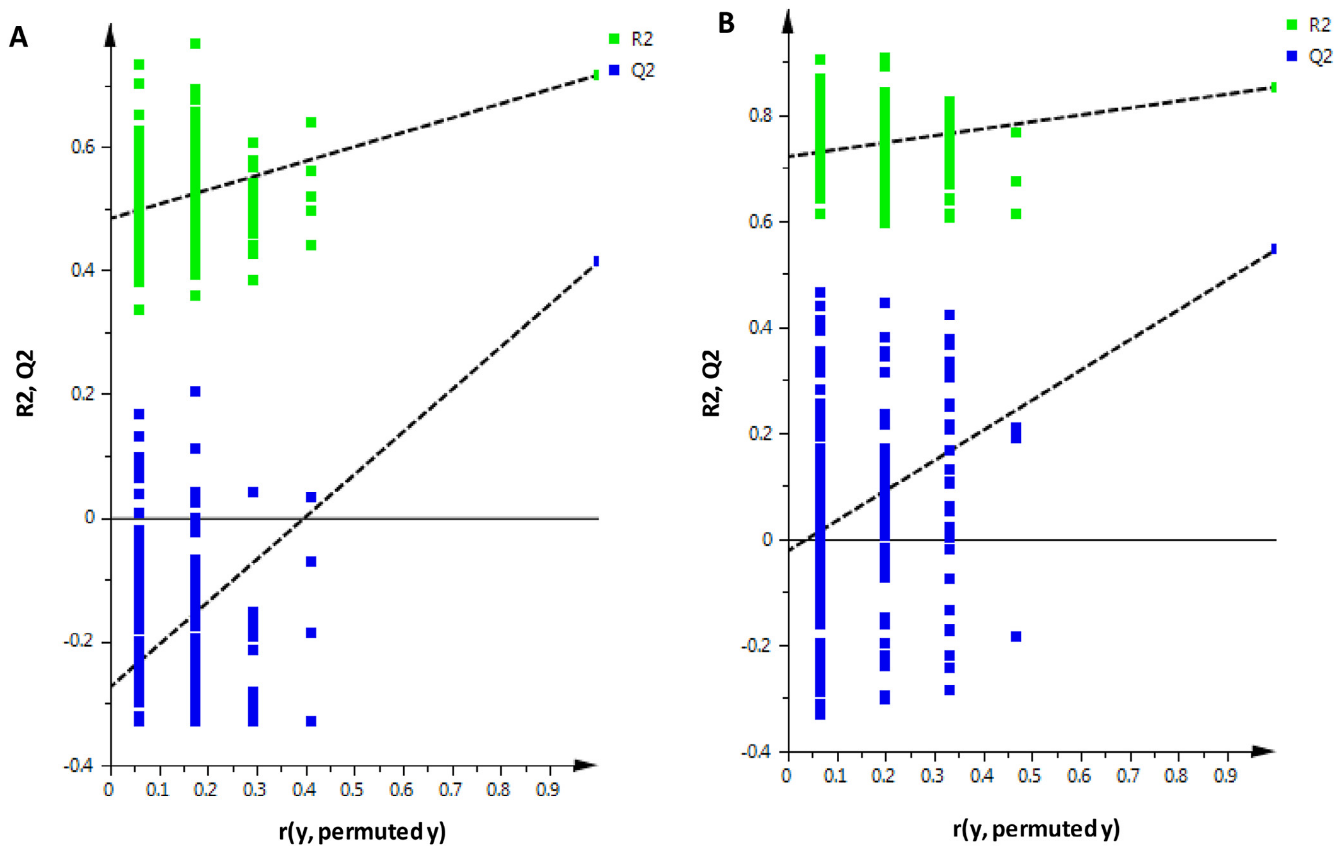

Fig. 2 – Validation plot of 100 permutation tests for OPLS-DA model built for plasma (A) and urine (B) after cranberry juice and after apple juice. Q^2-intercept of –0.3 or –0.08 is less than 0.05 indicating a valid model.

another way to visualize the 7-fold internal validation. Cross-validation and permutation test in many ways provide a reasonable approximation of the predictive ability of a PLS or OPLS model (Eriksson, 2006). External validation uses an independent set of test data to evaluate predictive performance of the PLS or OPLS model that are built on the training set; therefore it is a more scrupulous and demanding method (Eriksson, 2006). In the present study, to further test the predictability of the supervised models, we split the whole data set into a training set (approximate 70% of the samples) and test set (remaining 30% of the samples). Supervised models were constructed on the training data set and blindly classify the test set. This external validation procedure was repeated for 30 times, and correct classification rate was calculated by counting the correctly classified samples and divided by the total number of samples in the test set. Correct classification rate of 80.3% and 80.2% were obtained for plasma and urine samples, respectively (Table 2). Because the sample size was decreased in the training set compared to the whole data set, the correct classification rate obtained from the whole data set would be bigger than those calculated from the training set. The external validation results suggested that our statistic models were valid and could correctly classify unknown plasma and urine samples at a rate of no less than 80.2% and 80.3%, respectively. Both plasma and urine metabolomics analyses in this study suggested a differentiation between cranberry juice and apple juice consumption in young women.

3.5. *Metabolite identification*

To identify the contributing metabolites that are responsible for the separation of cranberry juice and apple juice consumption, a S-line™ technology, which is the tailored S-plot (Llorach, Urpi-Sarda, Jauregui, Monagas, & Andres-Lacueva, 2009; Wiklund et al., 2008) for NMR spectroscopy data, was used. It visualized both the modeled covariance and the modeled correlation structure between the X-variables and the predictive score in one graph. The p(ctr) is the centered loading vector of the first principal component. It was colored according to the absolute value of the correlation loading p(corr). A p(corr) > 0.5 was selected as significance level. The advantage of the S-line plot is that it displays the predictive loading in a form resembling the original NMR spectra. A list of makers detected in the S-line was then subjected to Welch's *t* test, and the *p*-value obtained for each marker was smaller than 0.01 (Table 3). Both multivariate and univariate analyses concluded eight significant metabolites were responsible for the separation between drinking cranberry juice and apple juice consumption. The contributing metabolites were identified by comparing their NMR spectra with published papers (Bouatra et al., 2013; Psychogios et al., 2011) and those registered in the Human Metabolome Database (www.hmdb.ca). These markers found from plasma and urine metabolomic data were summarized in Table 3.

A-type proanthocyanidins in cranberry juice or B-type proanthocyanidins in apple juice have very low absorption rate.

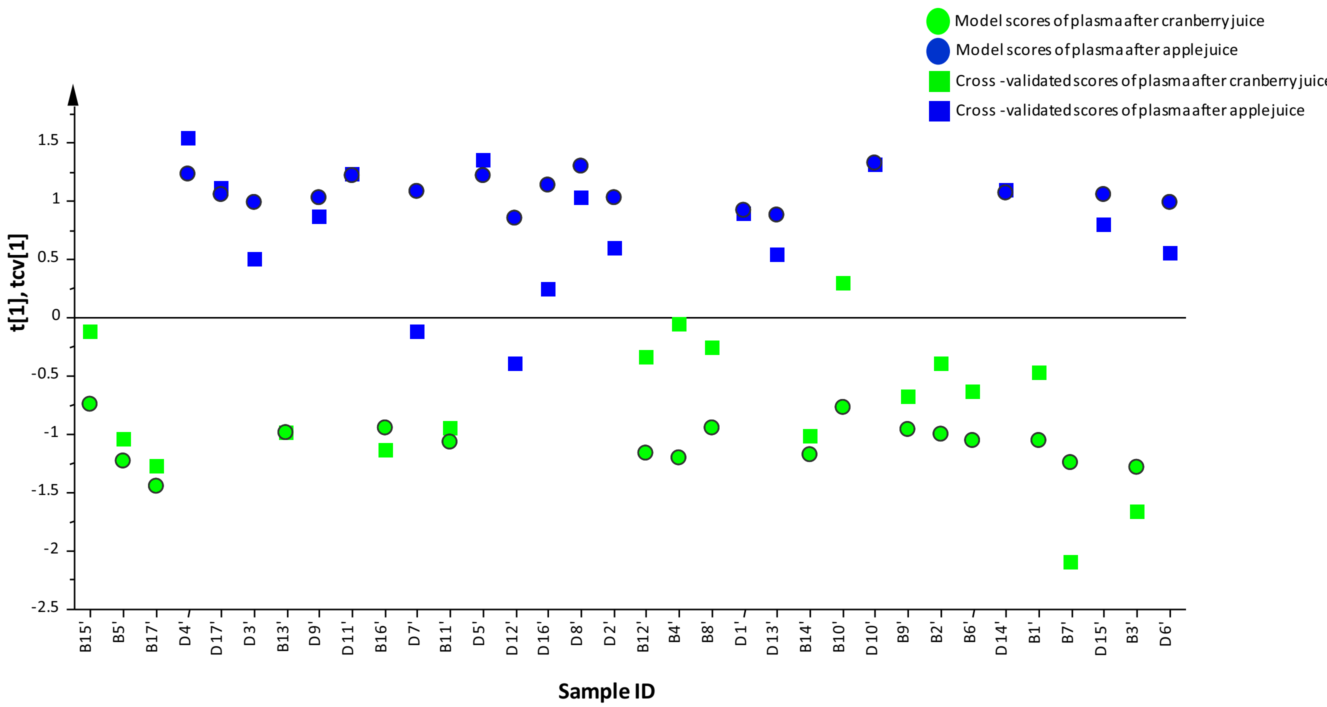

Fig. 3 – Model score plot (circles) and cross-validated score plot (squares) of OPLS-DA model for plasma following juice drinking. Green color: plasma after cranberry juice. Blue color: plasma after apple juice. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Only a small portion of epicatechin and proanthocyanidin oligomers with DP < 5 were absorbed in the small intestine (Ou & Gu, 2014). The absorption rate was below 5% (Gonthier et al., 2003; Rzeppa, Bittner, Döll, Dänicke, & Humpf, 2012; Tsang et al., 2005). The majority of proanthocyanidin oligomers and polymers reach the colon, where both A- and B-type proanthocyanidins are degraded by gut microflora to form various microbial metabolites. Part of the microbial metabolites were low molecular weight phenolic acids and phenylvalerolactones (Ou & Gu, 2014). In the present study, the detection of these microbial metabolites by NMR spectroscopy was limited due to their low levels in urine or plasma.

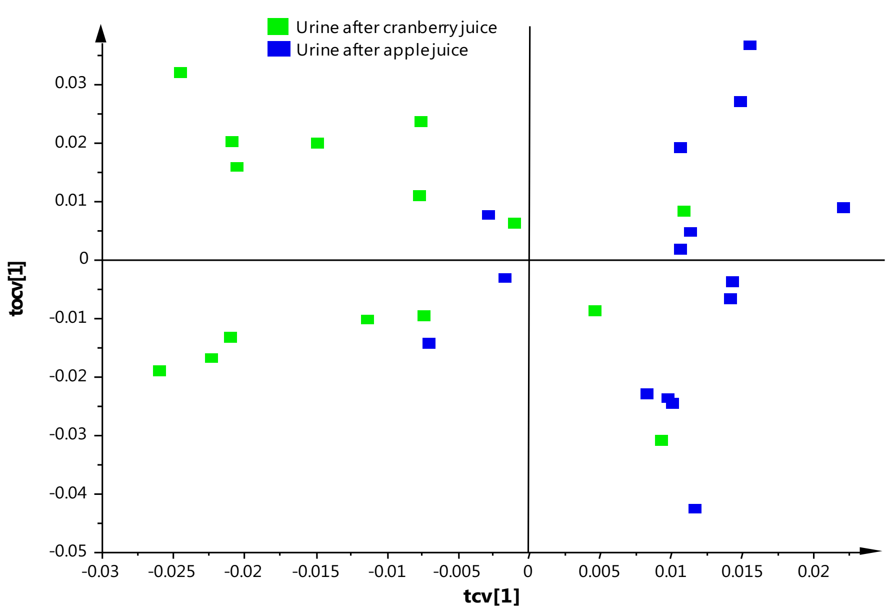

Fig. 4 – Cross-validated score plot of OPLS-DA model for urine following juice drinking. Green squares: urine after cranberry juice. Blue squares: urine after apple juice. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 3 – Summary of the metabolite profile changes in plasma and urine of young women after drinking cranberry juice and apple juice.

	Metabolites	Chemical shift (multiplicity)	*p*-value*	Cranberry juice *vs.* apple juice#
Plasma	Lactate	1.32 (d), 4.11 (q)	<0.01	↓
	D-glucose	3.22 (dd), 3.40 (q), 3.46 (m), 3.70 (m), 3.83 (m), 3.90 (m), 5.23 (d)	<0.01	↓
	Citric acid	2.52 (d), 2.62 (d)	<0.01	↑
	Unknown 1	2.36 (s)	<0.01	↓
	Unknown 2	4.01 (m)	<0.01	↓
	Unknown 3	3.56 (m)	<0.01	↓
Urine	Hippuric acid	3.96 (d), 7.54 (t),7.63 (t), 7.82 (d), 8.53 (br,s)	<0.01	↑
	Unknown 4	2.11 (s)	<0.01	↑

* p-value obtained from Welch's *t* test. Benjamini–Hochberg procedure was conducted to control false discoveries and conclude that all these variables are significantly different at $\alpha = 0.01$.
\# Arrows indicated a decrease or increase in metabolite level in plasma or urine after cranberry juice consumption compared to apple juice.

However, by using a global metabolomics approach we were able to find several metabolites that are responsible for the separation of cranberry juice and apple juice consumption, which were marked on the S-line in Fig. 5A and B. Cranberry juice and apple juice consumption had different impact on endogenous metabolites in urine and plasma. The plasma level of citric acid was considered to be increased after consumption of cranberry juice according to its loading profile. Although its relatively low variable magnitude makes it not an ideal case, its correlation loading p(corr) > 0.5 is accepted for being statistically significant. Lactate, D-glucose and three unidentified metabolites in plasma were higher after consumption of apple juice. The unidentified metabolite with chemical shift at 3.56 (m) ppm and 4.01 (m) ppm was first identified as quinic acid by matching its NMR spectrum with published data in Human Metabolome Database. We then spiked the plasma samples with pure quinic acid to disprove the identification. Cranberry juice consumption caused a stronger increase in urinary excretion of hippuric acid and one unidentified metabolite. Hippuric acid is formed by the conjugation of benzoic acid with glycine in the liver, and then excreted in urine. Production of hippuric acid is mainly from two routes. One is from the consumption of foods containing benzoic acid, the other one is from the metabolism of polyphenols into benzoic acid by the gut microflora (Walsh et al., 2007). Proanthocyanidins were degraded by the gut microflora into benzoic acid in the colon and benzoic acid was converted to hippuric acid in the liver (Rechner et al., 2002). Previous animal study showed that consumption of cranberry powder caused a strong increase in urinary excretion of hippuric acid. Its quantity in urine was higher than any other urinary phenolic acids (Prior et al., 2010). Citric acid is an intermediate in the citric acid cycle intermediates. Plasma level of citric acid was elevated after cranberry juice consumption, suggesting an increased oxidative energy metabolism. Reduction in plasma level of lactate suggested that cranberry juice consumption may be associated with anaerobic glycolysis reduction (Van Dorsten, Daykin, Mulder, & Van Duynhoven, 2006).

Box-and-whisker plots of signal intensity of these eight metabolites were used to display their differences in plasma or urine level following juice consumption (Fig. S4). The median intensity of lactate, glucose, unknown 1 (singlet at 2.36 ppm), and unknown 2 (multiplet at 4.01 ppm) in plasma following cranberry juice consumption were about two times lower than those after drinking apple juice. It was consistent with Welch's *t* test, confirming their significantly low level in plasma after drinking cranberry juice (Table 3). It is interesting to notice that the whiskers on the box plots of hippuric acid and unknown 4 (singlet 2.11 ppm) following apple juice were considerably smaller compared to those following cranberry juice, indicating that contents of hippuric acid and unknown 4 (singlet 2.11 ppm) were consistently low throughout all urine samples following apple juice consumption. The results were consistent with both univariate and multivariate analyses that these two metabolites had significantly higher quantities in subjects' urine after drinking cranberry juice.

4. Conclusions

In conclusion, our study showed that global ^1H NMR metabolomics is a very effective approach to differentiate metabolic impacts of cranberry juice consumption from those of apple juice. Cranberry juice consumption caused a higher plasma level of citric acid and urinary excretion of hippuric acid, while apple juice intake increased the plasma concentration of lactate and D-glucose. In addition, four unidentified metabolites were detected and considered as significant markers of cranberry juice consumption. Several health benefits were associated with consumption of cranberry juices; however, the mechanisms remain unclear. The metabolic differences observed in this study may help to explain the physiological activities of proanthocyanidin-rich cranberry juices.

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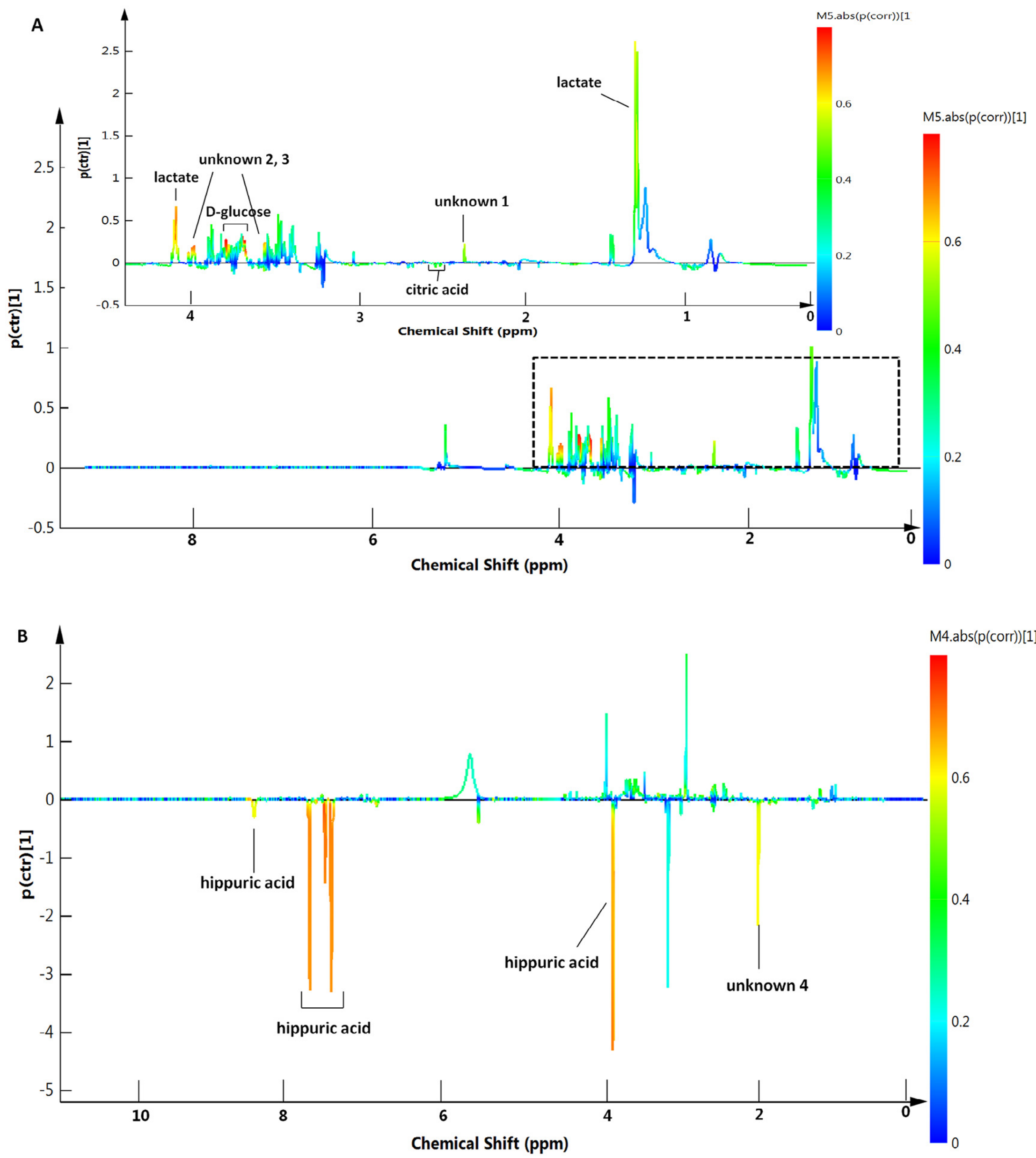

Fig. 5 – S-line associated with the OPLS score plots of data derived from plasma (A) and urine (B) after juice consumption. The x-axis is chemical shift derived from NMR spectra. The y-axis p(ctr)[1] is the centered loading vector of the first principal component. P(ctrl)[1] is colored according to the absolute value of the correlation loading p(corr). P(corr) > 0.5 is selected as significance level.

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Appendix: Supplementary material

Supplementary data to this article can be found online at doi:10.1016/j.jff.2015.01.018.

REFERENCES

Benjamini, Y., & Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, 57, 289–300.

Blatherwick, N. (1914). The specific role of foods in relation to the composition of the urine. *Archives of Internal Medicine*, 14, 409–450.

Bouatra, S., Aziat, F., Mandal, R., Guo, A. C., Wilson, M. R., Knox, C., Bjorndahl, T. C., Krishnamurthy, R., Saleem, F., & Liu, P.

(2013). The human urine metabolome. *PLoS ONE*, *8*, e73076.
Brindle, J. T., Antti, H., Holmes, E., Tranter, G., Nicholson, J. K., Bethell, H. W., Clarke, S., Schofield, P. M., McKilligin, E., & Mosedale, D. E. (2002). Rapid and noninvasive diagnosis of the presence and severity of coronary heart disease using ^{1}H-NMR-based metabonomics. *Nature Medicine*, *8*, 1439–1445.
Bylesjö, M., Rantalainen, M., Cloarec, O., Nicholson, J. K., Holmes, E., & Trygg, J. (2006). OPLS discriminant analysis: Combining the strengths of PLS-DA and SIMCA classification. *Journal of Chemometrics*, *20*, 341–351.
Chen, P. X., Bozzo, G. G., Freixas-Coutin, J. A., Marcone, M. F., Pauls, P. K., Tang, Y., Zhang, B., Liu, R., & Tsao, R. (2014). Free and conjugated phenolic compounds and their antioxidant activities in regular and non-darkening cranberry bean (*Phaseolus vulgaris L.*) seed coats. *Journal of Functional Foods*, in press.
Deziel, B. A., Patel, K., Neto, C., Gottschall-Pass, K., & Hurta, R. A. (2010). Proanthocyanidins from the American cranberry (*Vaccinium macrocarpon*) inhibit matrix metalloproteinase-2 and matrix metalloproteinase-9 activity in human prostate cancer cells via alterations in multiple cellular signalling pathways. *Journal of Cellular Biochemistry*, *111*, 742–754.
Dunn, W. B., Broadhurst, D. I., Atherton, H. J., Goodacre, R., & Griffin, J. L. (2011). Systems level studies of mammalian metabolomes: The roles of mass spectrometry and nuclear magnetic resonance spectroscopy. *Chemical Society Reviews*, *40*, 387–426.
Eriksson, L. (2006). *Multi-and megavariate data analysis* (2nd ed.). Sweden: MKS Umetrics AB.
Fonville, J. M., Richards, S. E., Barton, R. H., Boulange, C. L., Ebbels, T., Nicholson, J. K., Holmes, E., & Dumas, M. E. (2010). The evolution of partial least squares models and related chemometric approaches in metabonomics and metabolic phenotyping. *Journal of Chemometrics*, *24*, 636–649.
Foo, L. Y., Lu, Y., Howell, A. B., & Vorsa, N. (2000). A-type proanthocyanidin trimers from cranberry that inhibit adherence of uropathogenic P-Fimbriated *Escherichia coli*. *Journal of Natural Products*, *63*, 1225–1228.
Fuleki, T., Pelayo, E., & Palabay, R. B. (1994). Sugar composition of varietal juices produced from fresh and stored apples. *Journal of Agricultural and Food Chemistry*, 42, 1266–1275.
Giusti, M. M., & Wrolstad, R. E. (2001). Characterization and measurement of anthocyanins by UV-visible spectroscopy. In R. E. Wrolstad (Ed.), *Current protocols in food analytical chemistry (F1.2.1-F1.2.13)*. New Jersey: John Wiley & Sons, Inc.
Gonthier, M.-P., Donovan, J. L., Texier, O., Felgines, C., Remesy, C., & Scalbert, A. (2003). Metabolism of dietary procyanidins in rats. *Free Radical Biology and Medicine*, 35, 837–844.
Griffin, J. L. (2006). The Cinderella story of metabolic profiling: Does metabolomics get to go to the functional genomics ball? *Philosophical Transactions of the Royal Society B: Biological Sciences*, *361*, 147–161.
Gu, L., Kelm, M. A., Hammerstone, J. F., Beecher, G., Holden, J., Haytowitz, D., & Prior, R. L. (2003). Screening of foods containing proanthocyanidins and their structural characterization using LC-MS/MS and thiolytic degradation. *Journal of Agricultural and Food Chemistry*, 51, 7513–7521.
Howell, A. B., Reed, J. D., Krueger, C. G., Winterbottom, R., Cunningham, D. G., & Leahy, M. (2005). A-type cranberry proanthocyanidins and uropathogenic bacterial anti-adhesion activity. *Phytochemistry*, *66*, 2281–2291.
Kresty, L. A., Howell, A. B., & Baird, M. (2011). Cranberry proanthocyanidins mediate growth arrest of lung cancer cells through modulation of gene expression and rapid induction of apoptosis. *Molecules*, *16*, 2375–2390.
Lin, S., Chan, W., Li, J., & Cai, Z. (2010). Liquid chromatography/mass spectrometry for investigating the biochemical effects induced by aristolochic acid in rats: The plasma metabolome. *Rapid Communications in Mass Spectrometry*, *24*, 1312–1318.
Llorach, R., Garrido, I., Monagas, M., Urpi-Sarda, M., Tulipani, S., Bartolome, B., & Andres-Lacueva, C. (2010). Metabolomics study of human urinary metabolome modifications after intake of almond (*Prunus dulcis* (Mill.) DA Webb) skin polyphenols. *Journal of Proteome Research*, *9*, 5859–5867.
Llorach, R., Urpi-Sarda, M., Jauregui, O., Monagas, M., & Andres-Lacueva, C. (2009). An LC-MS-based metabolomics approach for exploring urinary metabolome modifications after cocoa consumption. *Journal of Proteome Research*, *8*, 5060–5068.
Manach, C., Hubert, J., Llorach, R., & Scalbert, A. (2009). The complex links between dietary phytochemicals and human health deciphered by metabolomics. *Molecular Nutrition & Food Research*, 53, 1303–1315.
Marchesi, J. R., Holmes, E., Khan, F., Kochhar, S., Scanlan, P., Shanahan, F., Wilson, I. D., & Wang, Y. (2007). Rapid and noninvasive metabonomic characterization of inflammatory bowel disease. *Journal of Proteome Research*, *6*, 546–551.
Nicholson, J. K., Lindon, J. C., & Holmes, E. (1999). 'Metabonomics': Understanding the metabolic responses of living systems to pathophysiological stimuli *via* multivariate statistical analysis of biological NMR spectroscopic data. *Xenobiotica*, *29*, 1181–1189.
Ou, K., & Gu, L. (2014). Absorption and metabolism of proanthocyanidins. *Journal of Functional Foods*, *7*, 43–53.
Prior, R. L., Lazarus, S. A., Cao, G., Muccitelli, H., & Hammerstone, J. F. (2001). Identification of procyanidins and anthocyanins in blueberries and cranberries (*Vaccinium spp.*) using high-performance liquid chromatography/mass spectrometry. *Journal of Agricultural and Food Chemistry*, *49*, 1270–1276.
Prior, R. L., Rogers, T. R., Khanal, R. C., Wilkes, S. E., Wu, X., & Howard, L. R. (2010). Urinary excretion of phenolic acids in rats fed cranberry. *Journal of Agricultural and Food Chemistry*, *58*, 3940–3949.
Psychogios, N., Hau, D. D., Peng, J., Guo, A. C., Mandal, R., Bouatra, S., Sinelnikov, I., Krishnamurthy, R., Eisner, R., & Gautam, B. (2011). The human serum metabolome. *PLoS ONE*, *6*, e16957.
Rechner, A. R., Kuhnle, G., Bremner, P., Hubbard, G. P., Moore, K. P., & Rice-Evans, C. A. (2002). The metabolic fate of dietary polyphenols in humans. *Free Radical Biology and Medicine*, *33*, 220–235.
Rzeppa, S., Bittner, K., Döll, S., Dänicke, S., & Humpf, H. U. (2012). Urinary excretion and metabolism of procyanidins in pigs. *Molecular Nutrition & Food Research*, *56*, 653–665.
Singleton, V., & Rossi, J., Jr. (1965). Colorimetry of total phenolics with phosphomolybdic-phosphotungstic acid reagents. *American Journal of Enology and Viticulture*, *16*, 144.
Tsang, C., Auger, C., Mullen, W., Bornet, A., Rouanet, J.-M., Crozier, A., & Teissedre, P.-L. (2005). The absorption, metabolism and excretion of flavan-3-ols and procyanidins following the ingestion of a grape seed extract by rats. *British Journal of Nutrition*, *94*, 170–181.
Van Dorsten, F. A., Daykin, C. A., Mulder, T. P., & Van Duynhoven, J. P. (2006). Metabonomics approach to determine metabolic differences between green tea and black tea consumption. *Journal of Agricultural and Food Chemistry*, *54*, 6929–6938.
Walsh, M. C., Brennan, L., Pujos-Guillot, E., Sébédio, J.-L., Scalbert, A., Fagan, A., Higgins, D. G., & Gibney, M. J. (2007). Influence of acute phytochemical intake on human urinary metabolomic profiles. *The American Journal of Clinical Nutrition*, *86*, 1687–1693.
Wiklund, S., Johansson, E., Sjöström, L., Mellerowicz, E. J., Edlund, U., Shockcor, J. P., Gottfries, J., Moritz, T., & Trygg, J. (2008). Visualization of GC-TOF-MS-based metabolomics data for identification of biochemically interesting compounds using OPLS class models. *Analytical Chemistry*, *80*, 115–122.

Yan, X., Murphy, B. T., Hammond, G. B., Vinson, J. A., & Neto, C. C. (2002). Antioxidant activities and antitumor screening of extracts from cranberry fruit (*Vaccinium macrocarpon*). *Journal of Agricultural and Food Chemistry*, 50, 5844–5849.

Zhao, S., Bomser, J., Joseph, E. L., & DiSilvestro, R. A. (2013). Intakes of apples or apple polyphenols decrease plasma values for oxidized low-density lipoprotein/beta$_2$-glycoprotein I complex. *Journal of Functional Foods*, 5, 493–497.